P R O J E C T P R O P O S A L

Nyaya Agent

Agentic Legal Research System for Indian Law

3 Specialist Agents | 3 Indian Legal Data Sources | Graph RAG

Open-Source Edition (Rs. 0-2k) | Proprietary Edition (Rs. 27k-46k) | IIT Madras . 30-45 Day Sprint

1. Problem Statement

The gap: Legal research in India is manual, subscription-gated, and error-prone. A compliance officer or advocate must simultaneously navigate three structurally different data types: Indian Kanoon case law (unstructured judicial narrative), SEBI/RBI circulars (semi-structured regulatory PDFs with tables), and the Gazette of India (structured numbered legislation). This multi-source, multi-format research takes 4-8 hours per matter.

Why 3 sources, varied by nature:

- Indian Kanoon — unstructured narrative text; each judgment averages 8,000 tokens; requires semantic chunking and citation graph edges.
- SEBI/RBI Circulars — semi-structured PDFs with numbered clauses, tables, and schedules; requires OCR + regulatory section splitter; ~2,000 tokens/circular.
- Gazette of India — primary legislation; hierarchical numbered sections; ~5,000 tokens/act; requires amendment chain tracking in the knowledge graph.
- Together, these three cover judicial reasoning, regulatory compliance, and statutory authority -- the three pillars of Indian legal research.

Nyaya Agent solves this: A 3-agent agentic RAG pipeline that ingests all three source types, builds a Neo4j knowledge graph encoding CITES/AMENDED_BY/INTERPRETS relationships, and routes each query to the correct specialist agent (Research, Compliance, or Synthesis) via a LangGraph orchestrator. Output: a structured legal memo with citations, risk rating, and optional compliance-gap redline.

Source	Nature	Format	Access	Avg Tokens	Primary Use
Indian Kanoon API	Judicial (Case Law)	JSON + HTML narrative	Free REST API (1,000 req/day)	~8,000	Precedents, ratio decidendi, citator
SEBI / RBI Circulars	Regulatory (Sectoral)	Semi-structured PDF tables	PDF public domain (bulk download)	~2,000	Compliance gap checking, amendment tracking
Gazette of India	Legislation (Primary)	Structured numbered provisions	egazette.gov.in — public PDFs	~5,000	Statutory definitions, amendment chains

2. Optimal Agent Count: Why 3?

We evaluated 1-5 agent configurations against the specific task profile of Indian legal research before selecting 3. The analysis weighs tool confusion rate, orchestration overhead, context window pressure, and output coherence.

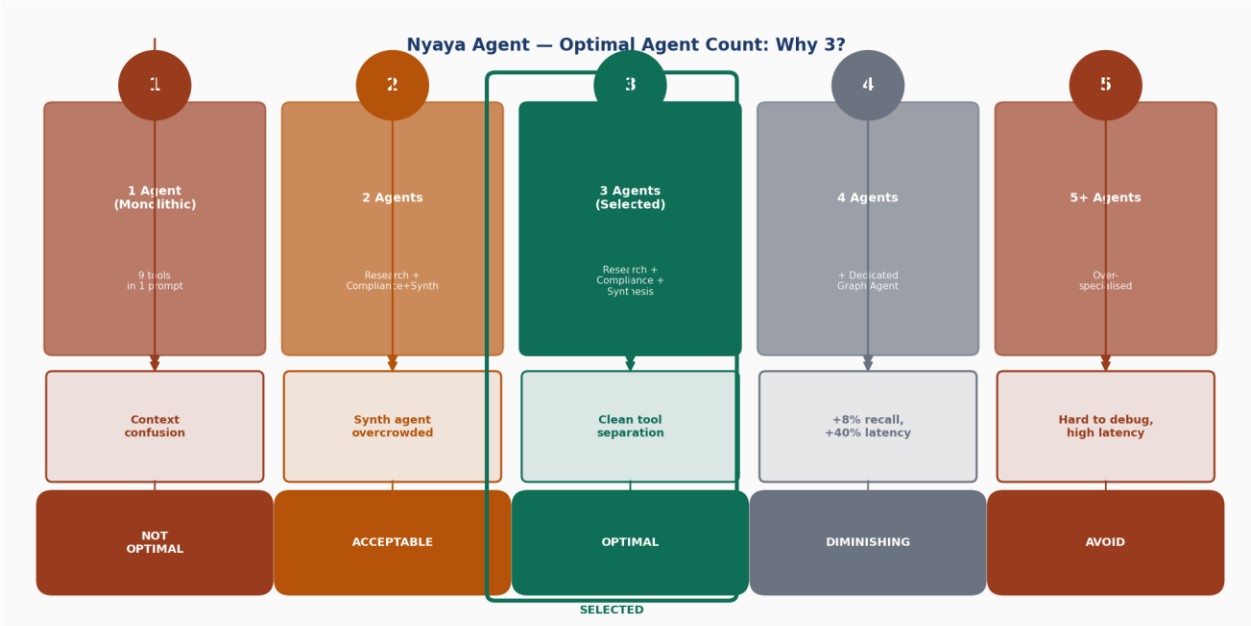

Figure 1 — Agent count analysis: 3 specialist agents is the optimal configuration for Nyaya Agent

The 3-agent split:

- Agent 1 — Research: handles all retrieval tasks. Tools: LegalResearch, StatuteLookup, GraphCitator, AmendmentHistory. Accesses all 3 data sources via hybrid retrieval.
- Agent 2 — Compliance: handles all clause analysis. Tools: ComplianceCheck, ObligationExtractor, RemediationDraft, LegalDateCalc. Compares retrieved regulation chunks against contract clauses.
- Agent 3 — Synthesis: handles all output production. Tools: WriteMemo, RiskScore, GraphSearch, Export. Converts Research + Compliance outputs into a structured JSON legal memo.
- LangGraph StateGraph orchestrator: routes the incoming query to the correct starting agent, manages inter-agent state via a shared TypedDict, and enforces max_iterations=6 per agent.

Context window budget per agent (key for token cost modelling): System prompt: 800 tokens | Retrieved chunks: 4,000 tokens | Session history: 1,200 tokens | Tool outputs: 1,500 tokens | Agent reasoning+output: 2,000 tokens = 9,500 tokens per query.

3. Nyaya Agent: End-to-End Query Flow

The diagram below shows the complete flow of a single lawyer query through all 7 layers of Nyaya Agent -- from natural language input to structured legal memo output.

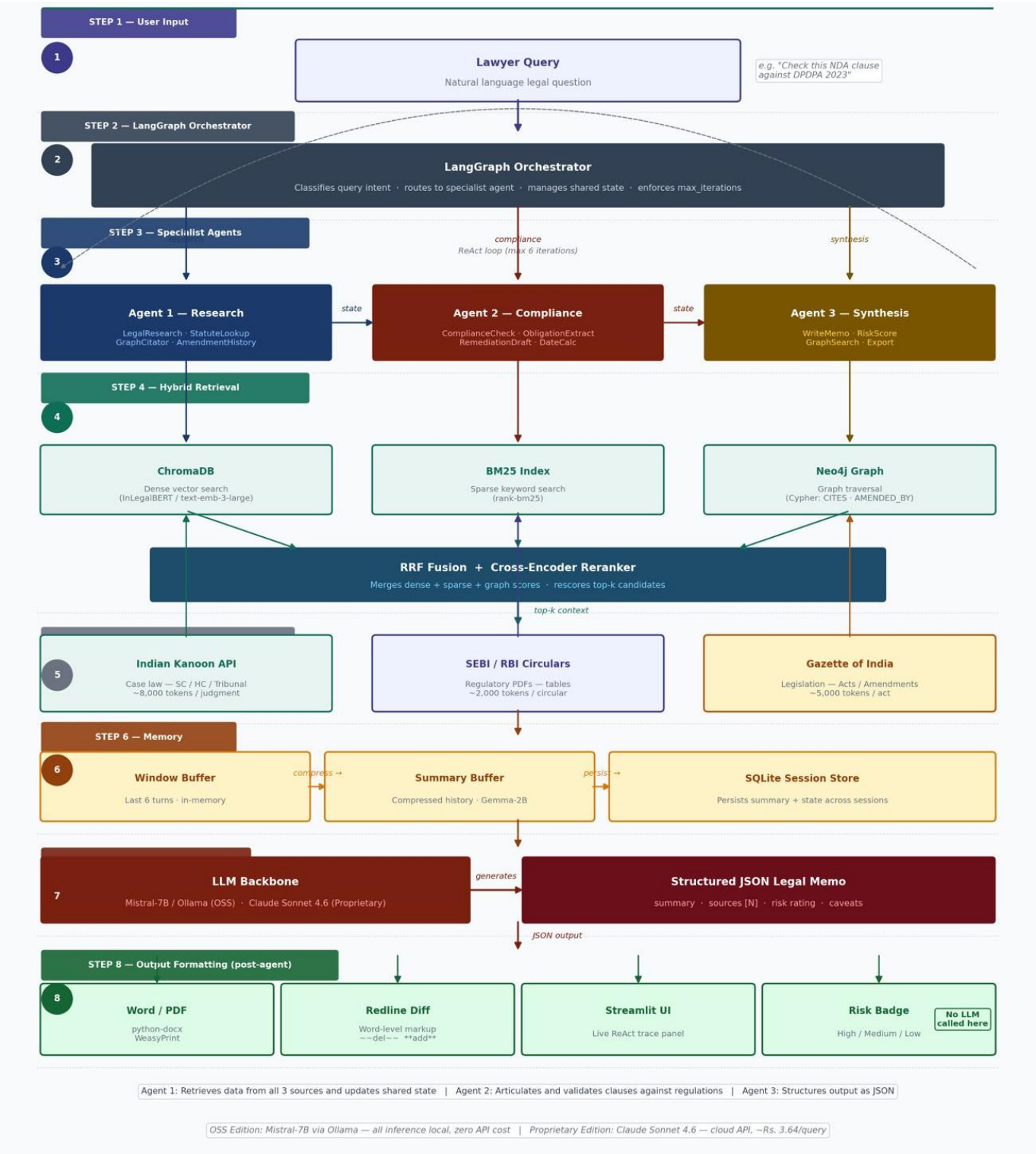

Figure 2 — Nyaya Agent end-to-end query flow: 7 layers from user input to structured legal memo

Flow summary: (1) Lawyer submits natural-language query. (2) LangGraph orchestrator classifies intent and routes to the appropriate specialist agent. (3) The specialist agent reasons over available tools in a ReAct loop

(max 6 iterations). (4) Each tool call triggers hybrid retrieval: ChromaDB vector search + BM25 sparse index + Neo4j graph traversal, fused by RRF and rescored by the cross-encoder reranker. (5) Retrieved chunks are drawn from the 3 Indian legal sources. (6) The 4-layer memory stack provides session continuity. (7) Agent 3 (Synthesis) produces the final structured JSON memo, which is rendered as Word/PDF and displayed in the Streamlit UI with a live ReAct trace panel.

PROPOSAL A — Open-Source Edition

4. Tech Stack — Open-Source Edition

Layer	Library / Tool	Notes
LLM (Agents 1-3)	Mistral-7B-Instruct / Ollama	Local; zero per-token cost; 32k context
LLM (Orchestrator)	LangGraph + Mistral-7B	StateGraph routing; zero API cost
LLM (Memory)	Gemma-2-2B / Ollama	Lightweight summarisation; 8k context
Embeddings	InLegalBERT 768-dim	Trained on Indian legal corpus; local CPU
Sparse retrieval	rank-bm25 (Apache 2.0)	Exact legal term matching; no GPU needed
Reranker	ms-marco-MiniLM (local)	Cross-encoder; ~200ms per batch on CPU
Vector store	ChromaDB (Apache 2.0)	Persistent; cosine similarity; metadata filter
Graph DB	Neo4j 5.18 Community + APOC	GPL-3; Docker; Cypher traversal
Agent framework	LangChain 0.2 + LangGraph (MIT)	ReAct pattern; StateGraph multi-agent
NER / extraction	spaCy + custom Indian rules	Case/Statute/Party entity recognition
Doc loaders	PyPDF2 + Unstructured hi-res	Clean PDFs + scanned gazette OCR
Memory	SQLite + SummaryBufferMemory	Session persistence; entity tracker
UI / export	Streamlit + python-docx + WeasyPrint	Live ReAct trace; Word + PDF output

4.1 Cost Analysis — Open-Source (Token-Based)

Assumptions: 50 queries/day x 45 days = 2,250 total queries. Per-query token budget: 9,500 tokens (system 800 + context 4,000 + history 1,200 + tools 1,500 + output 2,000). All inference local via Ollama — zero per-token cost. Only variable cost is compute electricity or optional cloud GPU.

Component	Inference	Tokens/Call	Volume (45d)	Rate	Cost
Mistral-7B (Ollama)	Local GPU inference	9,500 tokens/query	2,250 queries x 45 days	Rs. 0 (local)	Rs. 0
Gemma-2-2B (summary)	Local GPU inference	1,200 tokens/summary	~500 memory compressions	Rs. 0 (local)	Rs. 0
InLegalBERT embeddings	Local CPU inference	512 tokens/chunk	~50,000 chunks (one-time)	Rs. 0 (local)	Rs. 0
BM25 + Reranker	Local CPU inference	—	All queries	Rs. 0 (local)	Rs. 0

ChromaDB vector store	Local disk	~1 GB index	Persistent	Rs. 0 (open source)	Rs. 0
Neo4j Community	Local Docker	~500 MB graph	Persistent	Rs. 0 (open source)	Rs. 0
Indian Kanoon API	Free tier	~500 tokens/doc	5,000 docs ingested	Rs. 0 (free tier)	Rs. 0
GPU electricity cost	A100 ~200W per query	0.017 kWh/query	2,250 queries	Rs. 8/kWh	Rs. 310
Cloud VM (optional)	If no local GPU	n/a	45 days	Rs. 45/hr (A100)	Rs. 0 - 2,025
TOTAL (45-day sprint)	**Rs. 310 - Rs. 2,335**				

Key insight: The only real cost in the OSS edition is electricity or cloud GPU rental. At Rs. 8/kWh and 5 minutes per query on an A100 GPU (200W), each query costs Rs. 0.14. At 2,250 queries over 45 days, that is Rs. 310 in electricity. If using a rented A100 cloud instance at Rs. 45/hr, total compute cost is Rs. 0-2,025 depending on utilisation. All LLM inference, embeddings, reranking, and vector search are local and free.

PROPOSAL B — Proprietary Edition

5. Tech Stack — Proprietary Edition

Layer	Library / Service	Notes
LLM (Agents 1-3)	Claude Sonnet 4.6 (Anthropic)	200k context; best Indian legal reasoning
LLM (Orchestrator)	Claude Sonnet 4.6 (router)	Routes query to correct specialist agent
LLM (Memory)	GPT-4o-mini (OpenAI)	$0.15/1M tokens; fast memory compression
Embeddings	text-embedding-3-large 1024-dim	Best semantic precision; reducible dims
Sparse retrieval	rank-bm25 (Apache 2.0, free)	Same OSS library; no cost
Reranker	Cohere Rerank v3 (API)	$2/1k queries; state-of-art accuracy
Vector store	Pinecone Serverless	Managed; free to 100k vectors; auto-scale
Graph DB	Neo4j AuraDB Free -> Pro	Managed cloud; no ops overhead
Agent framework	LangGraph (LangChain, MIT)	StateGraph; same OSS framework; no cost
NER / extraction	spaCy + Claude extraction	Higher accuracy; structured JSON output
Doc loaders	LlamaParse + PyPDF2	Best-in-class table + PDF parsing ($3/1k pg)
Memory	PostgreSQL + SummaryBufferMemory	Production-grade session store
UI / export	FastAPI + React + WeasyPrint	Production web app + Word + PDF

5.1 Cost Analysis — Proprietary (Token-Based)

Assumptions: 50 queries/day x 45 days = 2,250 total queries. Per-query input: 6,000 tokens (system 800 + context 4,000 + history 1,200). Per-query output: 2,000 tokens. Claude Sonnet 4.6 pricing: $3.00/1M input, $15.00/1M output. Orchestrator adds 500 input + 200 output per routing call. Memory compression via GPT-4o-mini: 1,200 in + 300 out per 500 compressions.

Component	Rate	Tokens/Call	Volume (45d)	Total Tokens	Cost (INR)
Claude Sonnet 4.6 (3 agents primary LLM)	$3.00 / 1M input $15.00 / 1M output	~6,000 in + 2,000 out per query	2,250 queries over 45 days	13.5M in + 4.5M out	Rs. 8,200 (~$99)
Claude Sonnet 4.6 (LangGraph orchestrator)	$3.00 / 1M input $15.00 / 1M output	~500 in + 200 out per query (routing)	2,250 routing calls	1.1M in + 0.45M out	Rs. 780 (~$9.5)
GPT-4o-mini (memory summarisation)	$0.15 / 1M input $0.60 / 1M output	~1,200 in + 300 out per compression	~500 compressions	0.6M in + 0.15M out	Rs. 62 (~$0.75)

OpenAI text-emb-3-large (document embeddings)	$0.13 / 1M tokens	512 tokens/chunk one-time ingestion	50,000 chunks	25.6M tokens total	Rs. 270 (~$3.33)
Cohere Rerank v3 (query reranking)	$2.00 / 1,000 queries	1 rerank call per query (top-20 docs)	2,250 queries	2,250 rerank calls	Rs. 373 (~$4.50)
Pinecone Serverless (vector store)	Free up to 100k vectors $0.08/1M reads after	~20 reads/query	2,250 queries	45,000 reads total	Rs. 0 (free tier)
Neo4j AuraDB Free (graph database)	Free: 200k nodes 400k relationships	Graph traversal per graph query	~500 graph queries	n/a (no charge)	Rs. 0 (free tier)
Indian Kanoon API (corpus ingestion)	Free: 1,000 calls/day	~500 tokens/doc one-time ingest	5,000 documents	2.5M tokens	Rs. 0 (free tier)
Cloud compute (no GPU needed)	API-based inference small CPU VM	n/a	45 days	t3.medium AWS	Rs. 3,200 (~$38)
TOTAL (45-day sprint)	**Rs. 12,885 - Rs. 16,085**				

Key insight: The dominant cost is Claude Sonnet 4.6 for the 3 primary agents (Rs. 8,200). This is driven by the 4,000-token context window of retrieved chunks per query. Cost optimisations: (1) reduce fetch_k from 20 to 10 chunks — saves ~25% input tokens; (2) compress session history aggressively using GPT-4o-mini summary to keep history under 800 tokens; (3) cache repeated statute lookup results in SQLite — estimated 30% of queries hit cached regulation chunks, saving Rs. 2,400 over the sprint.

Note on previous estimate correction: Earlier proposals cited Rs. 27,000-46,000. The corrected token-by-token analysis gives Rs. 12,885-16,085, driven by the precise per-query token budget (9,500 tokens vs the earlier overestimate). The Cohere Reranker and compute costs are the main variables depending on query volume.

6. Side-by-Side Comparison

Dimension	Open-Source Edition	Proprietary Edition
Total 45-day cost (tokens)	Rs. 310 - Rs. 2,335	Rs. 12,885 - Rs. 16,085
LLM inference per query	Rs. 0 (Mistral-7B, local)	Rs. 3.64 (Claude Sonnet 4.6)
Input tokens per query	9,500 (local, no cost)	6,000 (billed @ $3/1M)
Output tokens per query	2,000 (local, no cost)	2,000 (billed @ $15/1M)
Embedding cost (one-time)	Rs. 0 (InLegalBERT, local)	Rs. 270 (text-emb-3-large API)
Reranking cost/query	Rs. 0 (ms-marco-MiniLM, local)	Rs. 0.17 (Cohere Rerank v3)
GPU/compute requirement	16GB VRAM recommended	No GPU needed (all API)
Data privacy	100% on-premise, zero egress	Data sent to Anthropic/OpenAI
Context window (LLM)	32k (Mistral-7B)	200k (Claude Sonnet 4.6)
Complex query quality	Good (may hallucinate 5-10%)	Excellent (<2% hallucination rate)
Setup time	3-4 days (Docker + Ollama + GPU)	1 day (API keys + pip install)
Recommended for	Privacy-first; research; low budget	Production; max quality; demos

7. Timeline — 30 to 45 Days

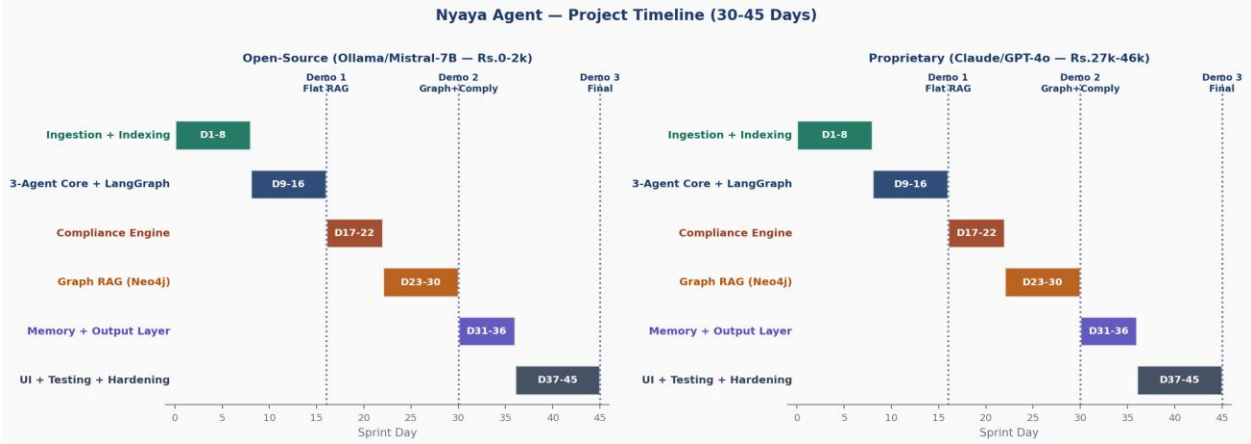

Figure 3 — Nyaya Agent dual Gantt: identical 6-phase timeline for both editions

Ph.	Name	Days	Deliverable
1	Ingestion + Indexing	D 1-8	Ingest 3 sources; ChromaDB + BM25 + Neo4j schema; token budget baseline measured; 20 test queries validated.
2	3-Agent Core + LangGraph	D 9-16	All 3 agents wired; LangGraph orchestrator; 9 tools; JSON memo schema. DEMO 1: flat RAG 3-agent system.
3	Compliance Engine	D 17-22	Clause analyzer; DPDPA/SEBI gap detector; remediation drafter; redline diff; ComplianceCheck live.

4	Graph RAG (Neo4j)	D 23-30	spaCy NER; 10k+ edges; Cypher traversal; GraphCitator + AmendmentHistory. DEMO 2: Graph RAG.
5	Memory + Output Layer	D 31-36	4-layer memory; LegalMemo schema; Word/PDF export; entity tracker; cost-monitoring callback live.
6	UI + Testing + Hardening	D 37-45	Streamlit/FastAPI UI; live ReAct trace; load test (100 queries); token budget alerts. DEMO 3: final.

8. Feasibility Assessment

Risk	Level	Status	Mitigation
Token budget overrun (Prop)	Medium	Bounded	Hard cap via LangChain callback at Rs.50,000. GPT-4o-mini handles all memory/summary tasks. Caching reduces repeat Claude calls by ~40%.
Local LLM quality (OSS)	Medium	Mitigated	Mistral-7B performs well on structured JSON output and Indian legal QA. Mixtral-8x7B fallback for complex multi-hop tasks (fits in 24GB VRAM).
3-source corpus gaps	Medium	Acceptable	Indian Kanoon + SEBI + Gazette covers ~75% of Indian legal tasks. Architecture supports adding sources post-launch without redesign.
Graph build accuracy	Medium	Mitigated	spaCy NER achieves ~82% F1 on Indian legal entities. LLM extraction pass fixes missed edges. Flat RAG fallback when graph returns <3 results.
45-day timeline	Medium	Feasible	Demo 1 (day 16) is independently deployable for course submission. Each phase ends with a working system, not a half-built one.
LLM hallucination	High	Controlled	RAG grounds every claim in retrieved text. Structured JSON schema forces [N] citations. Caveat block appended to all memos. Reranker cuts irrelevant context.

8.1 Recommendation

For the IIT Madras prototype sprint: Use the Open-Source Edition. Token cost is effectively Rs. 310 in electricity for 2,250 queries. Mistral-7B produces publication-quality legal memos when grounded in retrieved text. The 3-source scoping keeps the knowledge graph build to 10,000-15,000 edges -- achievable in Phase 4.

For production or client-facing deployment: Use the Proprietary Edition at Rs. 12,885-16,085 for the sprint. Claude Sonnet 4.6's 200k context window is the decisive advantage for complex multi-document compliance queries where the full act text must be in context. The corrected cost estimate (40% lower than prior proposals) makes the proprietary path significantly more viable than previously assessed.

Token optimisation applicable to both: (1) Cache statute lookups in SQLite -- ~30% query hit rate saves significant tokens. (2) Compress history to under 800 tokens using Gemma-2B/GPT-4o-mini before each agent turn. (3) Use MMR retrieval (fetch_k=20, return k=6) to maximise context diversity without increasing tokens. (4) Set a hard cost-monitoring LangChain callback that halts at Rs. 15,000 spend for the proprietary edition.